

Explainable Deep Learning Framework for Fake News Detection Using Reinforcement Learning-Based Adaptive Image-Text Fusion

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Declaration

I declare that this is my own work and this thesis does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other University or Institute of higher learning and, to the best of my knowledge and belief, it does not contain any material previously published or written by another person except where the acknowledgement is made in the text. I retain the right to use this content in whole or part in future works such as articles or books.

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The above candidate has carried out research for the undergraduate thesis under my supervision. I confirm that the declaration made above by the student is true and correct.

Name of Supervisor: Dr. Romesh Thanuja

Signature of the Supervisor:

Date:

Dedication

I dedicate this thesis to my family and friends, whose support, encouragement, and patience have been a constant source of strength throughout this research journey.

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I would like to express my sincere gratitude to my project supervisor, Dr. Romesh Thanuja, for his guidance, critical feedback, and encouragement throughout this research.

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Abstract

The rapid spread of fake news through social media has created a need for detection methods that can analyse both textual and visual evidence. Many online posts contain short text together with images, and the trustworthiness of each modality can vary across samples. Fixed multimodal fusion methods combine image and text predictions using predefined rules, but they do not explicitly adapt modality contribution when confidence, quality, or disagreement changes. This creates a research gap for explainable sample-level adaptive image-text fusion in fake news detection.

This research proposes an explainable deep learning framework for fake news detection using reinforcement learning-based adaptive image-text fusion. The framework trains separate image and text classifiers using a ResNet-based image branch and a DistilBERT-based text branch. It then constructs a reliability-aware state representation containing modality probabilities, confidence values, quality indicators, disagreement, confidence difference, and quality difference. A reinforcement learning fusion controller selects an action corresponding to image and text weights for each sample. The final prediction is produced by combining modality probabilities according to the selected action. The framework also provides explanation outputs through Grad-CAM image heatmaps, token-level text saliency, selected fusion actions, and modality weights.

The framework was evaluated on a Fakeddit-derived multimodal subset containing 26,471 available image-text samples across training, validation, and test splits. The image-only and text-only branches achieved test macro F1 scores of 0.7769 and 0.8388, respectively. The main adaptive fusion run achieved a test macro F1 of 0.8676, and validation-selected threshold tuning improved this to 0.8685. Paired McNemar tests showed that the difference between the selected RL run and strong fixed-fusion baselines was not statistically significant, while the RL controller significantly outperformed the matched supervised MLP fusion controller. The result is therefore interpreted as competitive adaptive fusion with additional sample-level interpretability through modality weights and explanation artifacts. The findings indicate that reliability-aware adaptive fusion is a useful direction for explainable multimodal fake news detection, while also highlighting the need for stronger training stability and broader external validation.

Keywords: Fake news detection; multimodal learning; reinforcement learning; adaptive fusion; explainable AI.

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List of Abbreviations

AI: Artificial Intelligence

BERT: Bidirectional Encoder Representations from Transformers

CNN: Convolutional Neural Network

DQN: Deep Q-Network

F1: F1 Score

Grad-CAM: Gradient-weighted Class Activation Mapping

MLP: Multi-Layer Perceptron

NLP: Natural Language Processing

ReLU: Rectified Linear Unit

RL: Reinforcement Learning

ROC-AUC: Receiver Operating Characteristic - Area Under the Curve

XAI: Explainable Artificial Intelligence

Chapter 1

Introduction

1.1 1.1 Introduction

Social media platforms have become major channels for the creation, consumption, and circulation of news. Their speed, openness, and low publication cost allow useful information to spread quickly, but the same properties also allow misleading and fabricated content to reach large audiences. Fake news detection has therefore become an important research problem in artificial intelligence because it requires automated methods that can analyse uncertain, noisy, and rapidly changing online content.

Many social media posts are multimodal. A post may contain a title or caption, an image, metadata, and user responses. Text can provide direct semantic evidence, while an image can provide contextual or visual evidence. However, the reliability of each modality is not constant. Some posts contain informative text but weak or irrelevant images, while others contain images that strongly influence the veracity decision. This thesis focuses on the problem of making the fusion of image and text evidence more adaptive and explainable.

1.2 1.2 Problem in Brief

The central problem addressed in this research is the lack of explainable, sample-level adaptive fusion in multimodal fake news detection. Conventional fixed fusion methods combine image and text predictions using a predefined rule, such as equal averaging or confidence-based weighting. These methods can be effective, but they do not explicitly learn a decision policy that selects different modality weights according to the reliability of each sample.

This limitation is important because fake news posts are heterogeneous. A single fusion rule may over-trust a weak modality in some samples and under-use a useful modality in others. In addition, many multimodal systems produce final predictions without explaining how much the image branch and text branch contributed to the decision. This weakens the practical usefulness of the model in settings where users need evidence, not only a label.

1.3 1.3 Background and Motivation

Prior research has shown that fake news detection is challenging because fake news can be linguistically persuasive, visually misleading, and context dependent. Early detection systems relied heavily on text, user features, or propagation patterns. More recent multimodal approaches combine textual and visual information because social media posts often use both modalities to shape user perception.

The motivation for this research comes from three observations. First, multimodal evidence is not equally reliable for every sample. Second, the fusion decision should be interpretable because fake news detection is a sensitive application. Third, reinforcement learning provides a natural way to model fusion as an action-selection problem, where the system observes the reliability state of a sample and selects a fusion action.

1.4 1.4 Novel Approach to Explainable Adaptive Image-Text Fusion

This research proposes an explainable deep learning framework that uses reinforcement learning-based adaptive fusion. The framework trains an image branch and a text branch separately, then uses their prediction probabilities, confidence scores, quality indicators, and disagreement features to form a reliability-aware state. A lightweight reinforcement learning controller selects one fusion action from a discrete action space. Each action corresponds to an image-text weight pair, and the final prediction is produced by applying the selected weights to the modality probabilities.

The proposed extension is novel in this project because it makes the fusion decision explicit. Instead of only reporting whether a post is real or fake, the framework records the selected action, image weight, text weight, final probability, and explanation artifacts. This provides a more inspectable decision process than a fixed fusion rule.

1.5 1.5 Research Aim

The aim of this research is to develop and evaluate an explainable multimodal fake news detection framework that adaptively fuses image and text predictions using a reinforcement learning-based reliability-aware fusion policy.

1.6 1.6 Research Objectives

The objectives of this research are:

1. To conduct a critical review of fake news detection, multimodal learning, adaptive fusion, reinforcement learning, and explainable AI methods.
2. To study the theoretical and technological foundations required for image-text fake news detection.
3. To design a reliability-aware adaptive fusion approach that selects image-text fusion weights at sample level.
4. To implement a reproducible framework containing image, text, reliability, fusion, baseline, robustness, and explainability components.
5. To evaluate the proposed framework using held-out data, baseline comparison, ablation, robustness analysis, and qualitative explainability evidence.

1.7 1.7 Research Questions

This research addresses the following questions:

1. How can image and text predictions be fused adaptively for multimodal fake news detection?
2. Which reliability features are useful for representing the state of a multimodal sample?
3. How does reinforcement learning-based adaptive fusion compare with image-only, text-only, fixed fusion, reliability-weighted fusion, and supervised fusion baselines?
4. How can the final prediction be explained at image, text, and fusion levels?
5. What limitations remain in terms of generalizability, stability, and reliability estimation?

1.8 1.8 Resource Requirements

The research required a Python-based deep learning environment, Google Colab GPU runtime, Google Drive storage, GitHub version control, and standard machine learning libraries. The main computational components were PyTorch, Transformers, scikit-learn, pandas, and image-processing utilities. The experiments used the Fakeddit multimodal dataset and a final computable subset for image-text evaluation.

1.9 1.9 Scope and Delimitations

The scope of this thesis is limited to binary fake news detection using image and text modalities. The work uses Fakeddit multimodal metadata and available downloaded images. It does not evaluate external social media streams, cross-domain datasets, user propagation graphs, or human participant studies. The explainability component provides qualitative image, text, and fusion-level explanations, together with a deletion-based faithfulness check on a 300-sample subset. The study does not claim complete explanation faithfulness or human-centered explanation validation.

1.10 1.10 Structure of the Thesis

Chapter 2 reviews developments and challenges in fake news detection and multimodal fusion. Chapter 3 explains the theoretical foundations of the technologies used in the framework. Chapter 4 presents the proposed reinforcement learning-based adaptive fusion approach. Chapter 5 describes the analysis and design of the framework. Chapter 6 explains implementation details. Chapter 7 evaluates the framework using experimental results, baseline comparisons, ablation, robustness, and explainability evidence. Chapter 8 concludes the thesis and discusses limitations and future work.

1.11 1.11 Summary

This chapter introduced the research problem, motivation, aim, objectives, research questions, scope, and thesis structure. The central gap is the need for explainable, sample-level adaptive image-text fusion in fake news detection. The next chapter reviews related work and positions this gap within existing fake news detection and multimodal learning research.

Chapter 2

Developments and Challenges in Multimodal Fake News Detection

2.1 Introduction

This chapter reviews prior work related to fake news detection, with emphasis on multimodal fake news detection and fusion strategies. The purpose of the review is to position the research gap addressed in this thesis: the need for explainable, sample-level adaptive image-text fusion. The chapter first discusses fake news detection as a social media analysis problem, then reviews text-based, image-based, and multimodal approaches. It also examines explainability and the limitations of existing fusion methods.

2.2 Fake News Detection in Social Media

Fake news detection has been studied as a data mining and artificial intelligence problem because online misinformation can be produced and distributed at high speed. Shu et al. described fake news detection on social media as a challenging task because the problem involves news content, user behaviour, propagation patterns, and psychological factors [1]. Their survey showed that fake news cannot be treated only as a conventional text classification problem, because social media content is shaped by both the message and the surrounding context.

In social media environments, fake news may be expressed through short captions, emotionally persuasive headlines, edited or misleading images, and posts designed to exploit existing beliefs. These properties make automated detection difficult. A model may learn superficial features from a particular dataset but fail when the event, topic, or presentation style changes. Therefore, recent research has increasingly focused on multimodal and generalizable detection methods.

2.3 2.3 Text-Based Fake News Detection

Text-based fake news detection uses textual features such as titles, captions, article content, writing style, and semantic representations. Earlier text-based systems used manually engineered linguistic features or classical machine learning methods. Later systems used neural language models to learn representations directly from text.

Transformer-based language models improved many natural language processing tasks by learning contextual word representations. BERT introduced deep bidirectional pretraining and showed that a pretrained language model could be fine-tuned for many downstream tasks with limited architectural modification [4]. DistilBERT reduced the computational cost of BERT through knowledge distillation while retaining much of its language understanding ability [5]. These models are relevant to fake news detection because they can represent short and noisy social media text more effectively than simple bag-of-words features.

However, text alone may be insufficient for multimodal posts. A caption may appear harmless while the image carries misleading context, or an image may be neutral while the text contains the false claim. Text models also risk overfitting to dataset-specific wording, topic patterns, or annotation artifacts. This limitation motivates multimodal approaches that use additional visual evidence.

2.4 2.4 Image-Based and Multimodal Fake News Detection

Visual evidence can provide important cues in fake news detection. Images may be manipulated, reused from unrelated events, or paired with misleading captions. Deep convolutional networks are commonly used for image representation. ResNet introduced residual learning to make deep visual networks easier to optimize and has become a widely used backbone for image classification [6].

Multimodal fake news detection combines text and image information. Wang et al. proposed Event Adversarial Neural Networks (EANN) for multimodal fake news detection and emphasized the problem of event-specific features that do not transfer well to unseen events [2]. Their work showed that multimodal representations can improve fake news detection, but it also highlighted the difficulty of learning robust and generalizable features.

The Fakeddit dataset was introduced as a large-scale multimodal fake news benchmark with multiple label granularities and image-text samples [3]. It is particularly relevant to this thesis because it provides image and text information suitable for evaluating multimodal fake news detection. The dataset also exposes a practical challenge: not all image

URLs remain available, so final multimodal experiments often require careful filtering and documentation of available samples.

2.5 2.5 Fusion Strategies in Multimodal Learning

Fusion is the process of combining information from multiple modalities. In multimodal fake news detection, fusion may occur at feature level, decision level, or through attention-based interaction. Feature-level fusion combines learned image and text embeddings before classification. Decision-level fusion combines modality predictions, such as probability scores. Attention-based and similarity-aware methods try to capture relationships between modalities.

SAFE introduced a similarity-aware multimodal fake news detection approach, showing that cross-modal similarity can be useful when analysing image-text consistency [9]. Progressive Fusion Networks later argued that multimodal fake news detection should not only use deep semantic features, because shallow-level visual and textual cues may also contain useful information [10]. TRIMOON focused on inconsistency-aware fusion and reported that image-text inconsistency can introduce fusion noise if it is not handled carefully [11]. These studies support the view that the relationship between modalities matters, not only the presence of multiple modalities.

Recent research has moved further toward adaptive and reliability-aware fusion. SAMPLE introduced similarity-aware multimodal prompt learning and used similarity-aware fusion to reduce noise from unrelated cross-modal representations [12]. MMFND used multi-task learning and cross-modal correlation weighting to guide multimodal fusion [13]. DAMMFND considered domain-aware multimodal decision-making and emphasized that modality contribution can vary across domains [14]. MSAF-Net, developed for short-video fake news detection, also used similarity-guided adaptive fusion to balance unimodal and inter-modal consistency features [15]. These recent methods show that adaptive fusion remains an active research direction.

Despite these advances, many fusion methods still provide limited sample-level explanation. A fixed averaging rule is simple but applies the same weighting behaviour to every sample. Confidence-weighted fusion is more flexible, but it remains deterministic and may not fully capture richer reliability features such as image quality, text quality, and modality disagreement. Deep feature-fusion models may learn complex interactions, but their fusion decisions can be difficult to inspect. The present research therefore focuses on a lightweight reinforcement learning controller that exposes the selected action and modality weights for each sample.

2.6 2.6 Explainability in Fake News Detection

Explainability is important in fake news detection because the output can influence user trust and decision-making. A prediction label alone may be insufficient when users need to understand why a post was classified as fake or real. In multimodal systems, explanation should ideally show evidence from image, text, and fusion levels.

Grad-CAM provides visual explanations for convolutional neural networks by highlighting image regions that influence a target prediction [8]. Token saliency methods can show which words or subword tokens influenced a text model. These explanation methods help inspect unimodal branches. However, multimodal fake news detection also needs fusion-level explanation: how much each modality contributed and why the final decision was formed.

Recent work has also connected fake news detection with interpretability and trust. Fusion-based approaches have been proposed to improve both detection and interpretability [16], and recent explainable multimodal fake-news studies show that robustness and transparency are becoming central concerns in misinformation detection [17]. The proposed research addresses this direction by exposing the selected fusion action, image weight, text weight, final probability, and branch-level saliency artifacts. This does not fully solve the broader problem of explanation faithfulness, but it improves transparency compared with a black-box final prediction.

2.7 2.7 Comparative Analysis of Existing Methods

Table 2.1 summarizes key categories of related work and their relevance to this research.

Table 2.1: Comparative analysis of related fake news detection methods

Method Category	Example Work	Strength	Limitation Relevant to This Thesis
Social media fake news surveys	Shu et al. [1]	Defines fake news detection challenges and datasets	Does not propose an adaptive multimodal fusion mechanism
Event-aware multimodal detection	EANN [2]	Uses text and image features and addresses event transfer	Fusion is not presented as an explicit sample-level action policy
Large multimodal benchmark	Fakeddit [3]	Provides large-scale image-text fake news data	Dataset availability and multimodal filtering must be handled carefully
Transformer text modelling	BERT [4], Distil-BERT [5]	Strong contextual text representation	Text-only evidence can overfit or miss visual misinformation
CNN image modelling	ResNet [6]	Strong visual representation backbone	Image-only evidence may be weak without text context
Similarity-aware multimodal fusion	SAFE [9], SAMPLE [12]	Considers image-text relation and cross-modal noise	Fusion decisions may still be difficult to inspect at sample level
Progressive and inconsistency-aware fusion	PFN [10], TRI-MOON [11]	Studies shallow/deep cues and modality inconsistency	Does not formulate fusion as an explicit action-selection policy
Recent adaptive/domain-aware fusion	MMFND [13], DAMMFND [14], MSAF-Net [15]	Shows that modality contribution and cross-modal similarity vary across settings	Often uses complex fusion modules that may reduce direct interpretability
Visual explainability	Grad-CAM [8]	Highlights image evidence	Does not explain text contribution or fusion action by itself

2.8 2.8 Research Gap

The literature indicates that multimodal fake news detection can benefit from both textual and visual evidence. It also shows that modality similarity, inconsistency, cross-modal noise, and domain variation are current research concerns. However, several gaps remain. First, many systems use fixed or opaque fusion methods that do not clearly adapt modality weights at sample level. Second, reliability information such as modality confidence, quality, and disagreement is not always explicitly represented in the fusion decision. Third, explanation often focuses on branch-level evidence but does not fully explain how modalities were combined.

Therefore, the research gap addressed in this thesis is the lack of an explainable, reliability-aware, sample-level adaptive fusion mechanism for image-text fake news detection. This thesis proposes a reinforcement learning-based fusion controller to address this gap by selecting a fusion action for each sample and exposing the resulting modality weights.

2.9 2.9 Summary

This chapter reviewed fake news detection, text-based modelling, image-based modelling, multimodal fusion, and explainability. The review showed that multimodal approaches are important but still face challenges in adaptive fusion and interpretability. The next chapter studies the theoretical foundations needed to design a reliability-aware reinforcement learning fusion framework.

Chapter 3

Theoretical Foundations of Reliability-Aware Adaptive Fusion

3.1 Introduction

This chapter presents the theoretical and technological foundations used in the proposed framework. The framework combines deep learning-based image classification, transformer-based text classification, reinforcement learning-based fusion, reliability estimation, and explainable AI. The chapter explains why these technologies were selected and how they support the research gap identified in Chapter 2.

3.2 Multimodal Representation Learning

Multimodal representation learning studies how information from different data types can be represented and combined. In fake news detection, the most common modalities are text and image. Text provides semantic information such as claims, captions, named entities, and emotional wording. Images provide visual evidence such as objects, scenes, screenshots, or manipulated content.

There are three broad fusion levels. Early fusion combines raw or low-level features before deeper processing. Intermediate fusion combines learned feature representations. Late fusion combines prediction scores or decisions from separate branches. This thesis uses a late-fusion design because the image and text branches can be trained and evaluated separately, and the fusion controller can operate on compact reliability features rather than high-dimensional embeddings.

The advantage of this design is interpretability. Since the controller receives explicit image and text probabilities, confidence values, and quality indicators, the selected fusion action can be traced back to understandable state features. The limitation is that the controller may not capture all fine-grained cross-modal semantic interactions that an end-to-end multimodal transformer might learn.

3.3 Transformer-Based Text Modelling

Transformer-based language models use self-attention mechanisms to learn contextual representations of tokens. BERT demonstrated the value of bidirectional pretraining by conditioning on both left and right context [4]. DistilBERT compressed BERT through knowledge distillation to produce a lighter model suitable for lower-resource settings [5].

This thesis uses DistilBERT as the text branch because the dataset contains short social media titles and captions, and contextual representation is useful for this type of text. The model receives cleaned text, encodes it using the pretrained language model, and produces a binary prediction probability. The text branch probability and confidence are later used by the fusion controller.

Theoretical strengths of transformer-based text modelling include contextual token representation, transfer learning from large corpora, and effective fine-tuning. Limitations include computational cost, sensitivity to dataset bias, and possible overfitting when the training data contains topic-specific cues.

3.4 Convolutional Image Modelling and Residual Networks

Convolutional neural networks learn hierarchical visual features from images. Lower layers often represent edges and textures, while deeper layers represent more task-specific patterns. ResNet introduced residual connections that help train deeper networks by allowing layers to learn residual functions [6].

This thesis uses a ResNet-based image branch because residual networks are reliable and widely used for image classification. The image branch receives downloaded images from the multimodal dataset and produces a binary prediction probability. This probability, together with confidence and image quality features, becomes part of the fusion state.

The image branch is important because visual content can contain evidence not available in text. However, image-only detection can be difficult when the image is ambiguous, irrelevant, low quality, or only meaningful when interpreted with the text. This supports the need for adaptive fusion rather than unimodal prediction.

3.5 3.5 Reinforcement Learning and Action-Selection Policies

Reinforcement learning studies how an agent can learn to select actions based on rewards. Deep reinforcement learning combines reinforcement learning with neural networks to approximate action-value functions or policies. Mnih et al. showed that a deep Q-network could learn control policies from high-dimensional inputs [7]. In this thesis, the input is not a raw visual state but a compact reliability-aware state.

The proposed fusion controller treats each sample as a state. The action corresponds to a selected image-text weight pair. The reward is based on whether the final fused prediction is correct. This formulation is appropriate because the task is not only to classify a sample but to decide which fusion behaviour should be used for that sample.

The reinforcement learning formulation provides two benefits. First, it makes fusion an explicit decision process. Second, the selected action can be inspected after prediction. The main limitation is that reinforcement learning can be sensitive to training configuration, reward design, and random seed, especially when the performance difference between actions is small.

3.6 3.6 Reliability Estimation for Image and Text Modalities

Reliability estimation attempts to measure whether a modality should be trusted for a particular sample. This thesis uses engineered reliability features rather than ground-truth reliability labels. For the image modality, quality indicators include features related to blur, contrast, entropy, and related image statistics. For the text modality, quality indicators are based on textual properties such as length and availability.

Confidence is another reliability signal. A prediction probability close to 0 or 1 indicates higher model confidence, while a probability near 0.5 indicates uncertainty. Disagreement between image and text predictions is also important because a sample where both branches agree may require less complex fusion than a sample where they conflict.

These reliability features are not perfect measures of truth. They are heuristic signals designed to support the fusion controller. The evaluation therefore includes ablation and policy analysis to test whether the reliability-aware state improves performance and produces meaningful weight patterns.

3.7 3.7 Explainable AI for Multimodal Decisions

Explainable AI aims to make model decisions more understandable to users and developers. In image classification, Grad-CAM uses gradients flowing into convolutional layers to produce heatmaps that highlight influential visual regions [8]. In text classification, token saliency can identify tokens that contributed strongly to the model output.

For multimodal fake news detection, explanation should include both branch-level and fusion-level information. Branch-level explanations show what influenced the image and text classifiers. Fusion-level explanations show how the modalities were combined. The proposed framework therefore records selected action, image weight, text weight, final probability, and final prediction together with branch-level explanation artifacts.

This approach improves transparency but does not fully prove explanation faithfulness. A future extension should evaluate explanations using perturbation tests, user studies, or quantitative faithfulness metrics.

3.8 3.8 Summary

This chapter explained the theoretical foundations of the proposed framework: multimodal representation learning, transformer-based text modelling, residual image modelling, reinforcement learning, reliability estimation, and explainable AI. These foundations support the proposed reliability-aware adaptive fusion mechanism described in Chapter 4.

Chapter 4

RL-Based Adaptive Image-Text Fusion Approach

4.1 Introduction

This chapter presents the conceptual approach proposed in this research. The purpose of the approach is to extend multimodal fake news detection by replacing a fixed image-text fusion rule with an adaptive and explainable fusion policy. The approach treats fusion as an action-selection problem in which a reinforcement learning controller observes reliability-related features and selects how much influence should be assigned to the image and text branches.

4.2 Hypothesis and Inspiration

The hypothesis of this research is that a reliability-aware adaptive fusion policy can provide competitive fake news detection performance while making the modality contribution more explainable at sample level. The approach is inspired by the observation that image and text evidence are not equally reliable for every social media post. A fixed fusion method applies the same rule even when one modality is uncertain, noisy, or in conflict with the other modality. An adaptive policy can instead select a fusion action based on the observed reliability state of the sample.

4.3 Inputs and Outputs of the Proposed Extension

The inputs to the proposed extension are the outputs of the image and text branches, together with reliability indicators derived from the sample. The image branch produces an image probability and an image confidence value. The text branch produces a text probability and a text confidence value. Additional reliability indicators represent image quality, text quality, modality disagreement, confidence difference, and quality difference.

The main output is a binary fake news prediction using the label standard 0 for real

and 1 for fake. The framework also outputs the final probability, selected fusion action, image weight, text weight, and explanation artifacts. These outputs make the decision more inspectable than a prediction label alone.

4.4 4.4 Process Workflow of the Proposed Extension

The proposed workflow contains six conceptual stages, as illustrated in Figure 4.1. First, the dataset is prepared and labels are standardized. Second, the image and text branches are trained separately. Third, both branches generate prediction probabilities for each split. Fourth, reliability features are computed and merged with modality predictions. Fifth, the reinforcement learning fusion controller selects a fusion action for each sample. Sixth, the framework generates classification metrics, policy analysis, robustness results, and explanation outputs.

The process is designed to separate unimodal representation learning from adaptive decision fusion. This separation allows the fusion controller to remain lightweight, because it uses a compact reliability-aware state instead of processing raw images or raw text directly.

The architectural novelty is the replacement of a fixed post-hoc fusion rule with a sample-level controller that observes reliability evidence before assigning modality weights. The novelty comparison diagram in this chapter makes this distinction explicit by separating simple fixed fusion from the proposed reliability-aware action-selection path.

4.5 4.5 Reliability-Aware State Representation

The reliability-aware state is the main input to the fusion controller. It contains:

1. Image prediction probability.
2. Image confidence.
3. Image quality.
4. Text prediction probability.
5. Text confidence.
6. Text quality.
7. Modality disagreement.
8. Confidence difference.
9. Quality difference.

This state representation was selected to capture both branch-level prediction strength and cross-modal conflict. The image and text probabilities represent the direct unimodal

evidence. The confidence features represent prediction certainty. The quality features provide engineered reliability cues. Disagreement and difference features help the controller identify samples where the modalities may require more careful balancing.

4.6 Adaptive Fusion Action Space

The reinforcement learning controller selects from a discrete set of fusion actions. Each action maps to a fixed pair of image and text weights. This design makes the policy easy to interpret because each selected action can be directly translated into a modality trust decision. A continuous action space may provide finer control, but the discrete action space was selected for simplicity, stability, and explainability.

4.7 Reward Design

The reward is based on whether the fused prediction is correct. If the selected fusion action leads to a correct prediction, the controller receives a positive reward. If the prediction is incorrect, it receives a negative reward. This binary reward directly connects the policy to classification correctness.

The reward design is intentionally simple. It supports clear interpretation but does not distinguish between highly confident and marginally correct predictions. A margin-based or log-loss-based reward is therefore identified as a possible future improvement.

4.8 Explanation Strategy

The framework provides explanation at three levels. At the image level, Grad-CAM heatmaps highlight visual regions that contributed to the image prediction. At the text level, token saliency identifies influential tokens from the text branch. At the fusion level, the selected action, image weight, text weight, final probability, and final prediction explain how the modalities were combined.

This explanation strategy is aligned with the research problem because it does not stop at model performance. It also provides evidence about why the final prediction was formed in a particular way.

4.9 4.9 Positioning within the AI Body of Knowledge

The proposed extension lies at the intersection of multimodal learning, reinforcement learning, and explainable AI. The image and text branches provide unimodal deep learning representations. The fusion controller applies reinforcement learning to select a decision action based on a compact state. The explanation layer supports interpretability by exposing modality weights and branch-level saliency.

The contribution is not the invention of a new image encoder or language model. Instead, the contribution is the use of a reliability-aware reinforcement learning controller to make image-text fusion adaptive and inspectable for fake news detection.

4.10 4.10 Summary

This chapter presented the proposed RL-based adaptive image-text fusion approach. The approach uses separate image and text branches, a reliability-aware state representation, a discrete fusion action space, and explanation outputs. The next chapter converts this conceptual approach into a system-level analysis and design.

The image and text branches are represented by functions that output fake-class probabilities:

$$p_i = f_i(x_i), \quad p_t = f_t(x_t) \quad (4.1)$$

The confidence of each modality is computed from the stronger binary-class probability:

$$c_m = \max(p_m, 1 - p_m), \quad m \in \{i, t\} \quad (4.2)$$

The reliability-aware state used by the fusion controller is defined as:

$$s = [p_i, c_i, q_i, p_t, c_t, q_t, |p_i - p_t|, c_i - c_t, q_i - q_t] \quad (4.3)$$

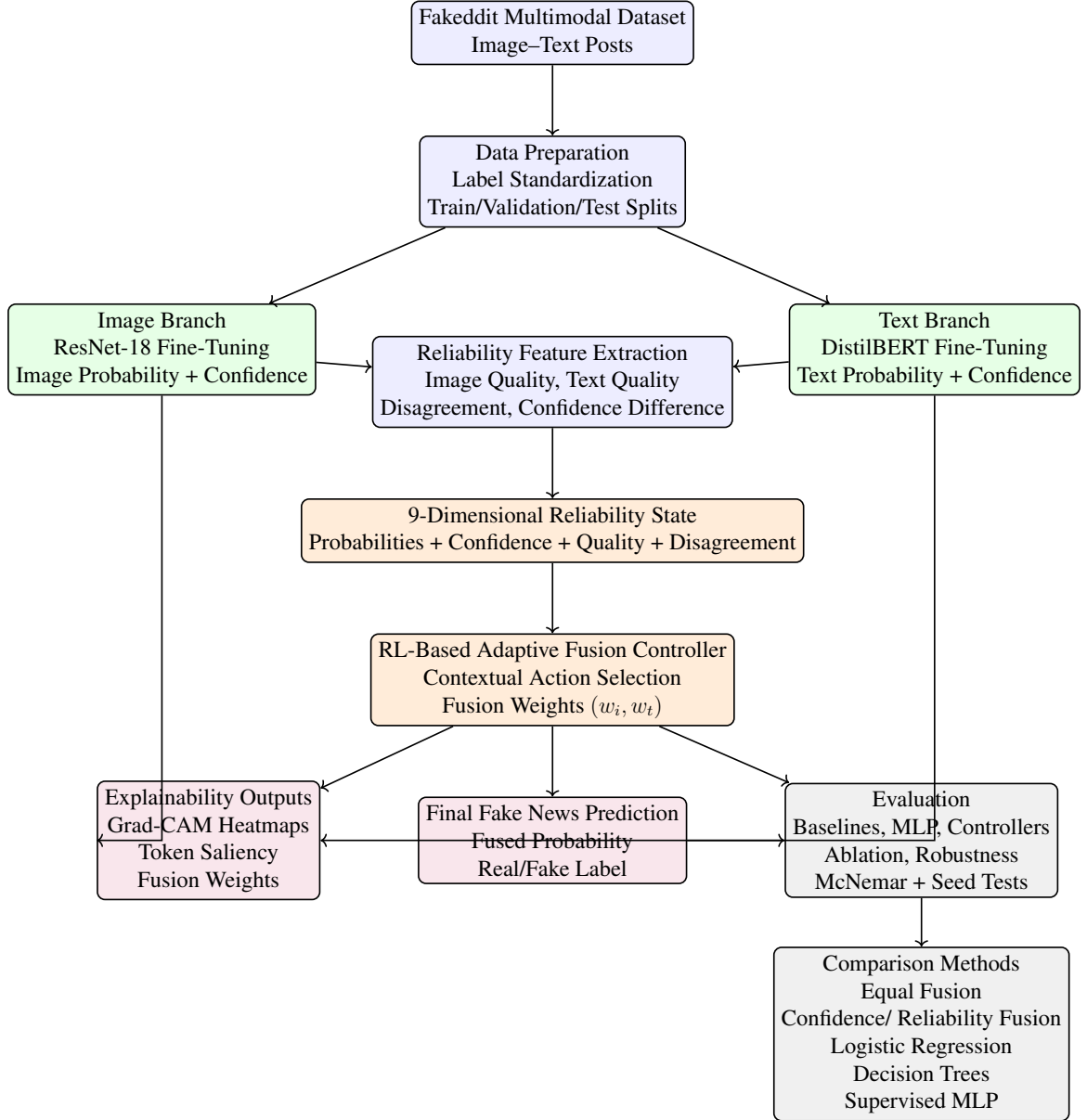
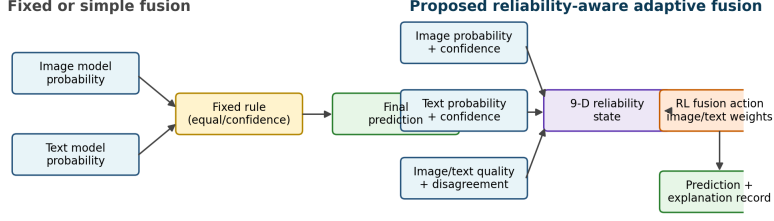


Figure 4.1: Overall architecture of the proposed explainable reinforcement learning-based adaptive image-text fusion framework for fake news detection.



Novelty: fusion is selected per sample from reliability evidence, and the selected action is inspectable.

Figure 4.2: Structural difference between fixed fusion and the proposed reliability-aware adaptive fusion controller.

Each discrete action maps to an interpretable image-text weighting pair:

$$\mathcal{A} = \{(0.90, 0.10), (0.75, 0.25), (0.60, 0.40), (0.50, 0.50), (0.40, 0.60), (0.25, 0.75), (0.10, 0.90)\} \quad (4.4)$$

$$a_k \mapsto (w_i^{(k)}, w_t^{(k)}), \quad w_i^{(k)} + w_t^{(k)} = 1 \quad (4.5)$$

The final fake-class probability is then produced by weighted late fusion:

$$p_f = w_i^{(k)} p_i + w_t^{(k)} p_t \quad (4.6)$$

The binary decision rule is:

$$\hat{y} = \begin{cases} 1, & p_f \geq \tau \\ 0, & p_f < \tau \end{cases} \quad (4.7)$$

The reward function links the selected fusion action directly to classification correctness:

$$r(s, a_k) = \begin{cases} +1, & \hat{y} = y \\ -1, & \hat{y} \neq y \end{cases} \quad (4.8)$$

Chapter 5

Analysis and Design of the Adaptive Fusion Framework

5.1 Introduction

This chapter converts the conceptual approach described in Chapter 4 into a system-level design. The chapter focuses on architecture, data flow, component responsibilities, and the positioning of the adaptive fusion extension. In accordance with the thesis guideline, this chapter does not present implementation-level code or algorithmic details; those are discussed in Chapter 6.

5.2 System Requirements and Assumptions

The framework was designed to satisfy the following functional requirements:

1. Prepare a paired image-text fake news dataset with standardized labels.
2. Train and evaluate a separate image classification branch.
3. Train and evaluate a separate text classification branch.
4. Generate modality probabilities and confidence scores for train, validation, and test splits.
5. Estimate image and text reliability features.
6. Train an adaptive fusion controller that selects sample-level modality weights.
7. Compare the proposed fusion model with deterministic and supervised baselines.
8. Generate policy analysis, robustness analysis, and explainability artifacts.
9. Export thesis-ready result summaries.

The framework assumes that each final evaluation sample contains both text and an available image. It also assumes that training, validation, and test splits remain separate throughout the pipeline, with validation used for model selection and threshold tuning, and test used only for final evaluation.

5.3 5.3 Top-Level Architecture

The top-level architecture, shown in Figure 5.1, contains six major layers: data preparation, unimodal modelling, reliability construction, adaptive fusion, evaluation, and explainability. The data preparation layer standardizes raw Fakeddit metadata and creates train, validation, and test splits. The unimodal modelling layer contains the image and text branches. The reliability construction layer merges modality predictions with confidence and quality indicators. The adaptive fusion layer uses the reliability-aware state to select image-text weights. The evaluation layer compares results across baselines and proposed models. The explainability layer generates image, text, and fusion-level explanations.

The architecture is modular so that each stage can be tested and rerun independently. This is important because image downloading, GPU training, and evaluation can fail or require reruns in a cloud environment.

5.4 5.4 Data Flow and Interaction Design

The data flow begins with raw Fakeddit metadata files. These files are converted into a standard schema containing sample identifier, image path or URL, text, and label. A balanced final subset is created for computation. The image downloader attempts to obtain image files and then creates image-available splits containing only samples with successfully downloaded images.

After the final available splits are created, the image branch and text branch are trained separately. Each branch then produces prediction outputs for train, validation, and test data. These outputs are merged into reliability files. The fusion controller uses the train reliability file for learning, the validation reliability file for model selection, and the test reliability file for final evaluation.

The evaluation outputs are written as JSON and CSV files so that the thesis can cite file-backed evidence. The notebooks provide a human-readable execution record for demonstration and viva purposes.

5.5 5.5 Design of Unimodal Branches

The image branch is designed as a visual classifier that maps an input image to a binary probability. A residual convolutional architecture is used because it provides a strong general-purpose image representation. The output of the image branch is not only a predicted class but also a probability and confidence value.

The text branch is designed as a transformer-based classifier that maps cleaned text into

a binary probability. It provides a text prediction, text probability, and text confidence. The text branch is kept separate from the image branch so that each modality can be evaluated independently before fusion.

This branch separation supports transparent evaluation. If the final fusion model improves over unimodal branches, the source of improvement can be examined through baseline and ablation results.

5.6 5.6 Design of the Adaptive Fusion Controller

The adaptive fusion controller is designed as the main research extension. It receives a compact reliability-aware state rather than raw image and text inputs. This state includes modality probabilities, confidence indicators, quality indicators, disagreement, and difference features. The controller selects a discrete fusion action, and each action maps to a specific pair of image and text weights.

This design makes fusion explainable because the selected action can be reported for each sample. The final prediction can therefore be explained in terms of the image branch probability, text branch probability, selected image weight, selected text weight, and final fused probability.

5.7 5.7 Design of Baseline and Comparison Components

The framework includes baseline components to make evaluation meaningful. Image-only and text-only baselines test the strength of each modality. Equal fusion and confidence-weighted fusion test simple decision-level multimodal fusion. Reliability-weighted fusion tests a deterministic use of reliability features. Supervised MLP fusion provides a learned non-RL comparison. These baselines are necessary because the proposed RL controller must be compared against simple and strong alternatives, not only against unimodal branches.

The comparison design also includes ablation experiments. Ablation allows the effect of state features to be analysed by comparing probabilities-only, prediction-confidence, and full reliability-aware states.

5.8 5.8 Design of Explainability Outputs

The explanation design has three levels. The image explanation uses a heatmap to indicate visually influential regions. The text explanation uses token saliency to indicate influential

words or subword tokens. The fusion explanation records selected action, modality weights, and final probability.

This design is aligned with the research problem because the model is expected to explain not only what it predicted, but also how image and text evidence were combined.

5.9 5.9 Summary

This chapter described the system design of the adaptive fusion framework. The design separates data preparation, unimodal modelling, reliability construction, adaptive fusion, evaluation, and explainability into modular components. The next chapter explains how this design was implemented in the project repository.

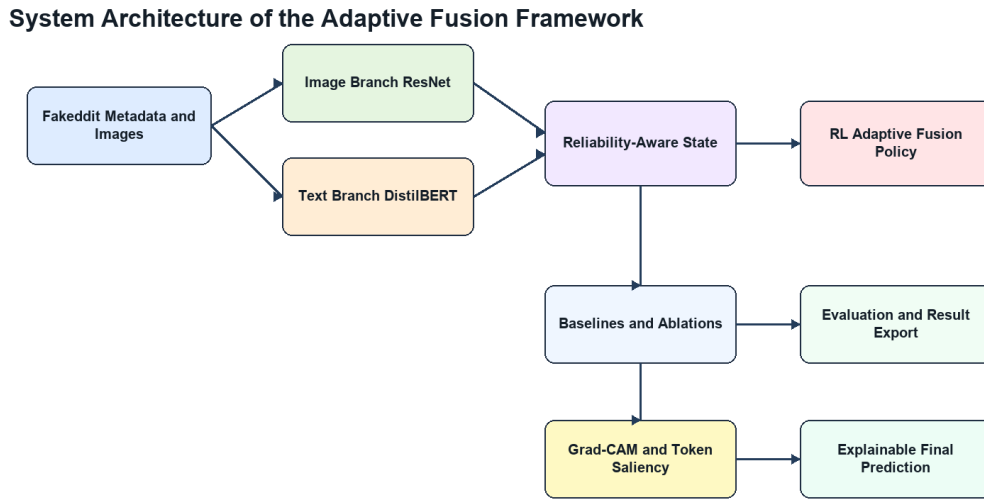
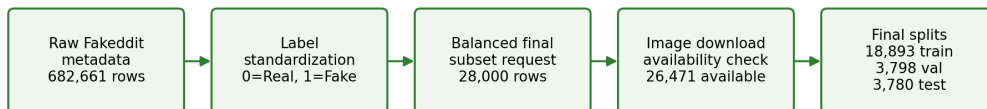


Figure 5.1: System architecture of the adaptive fusion framework.



Retention evidence: failed image downloads were recorded, not silently filled.

Download summary: requested 28,000; downloaded 26,471; failed 1,529.

Validation checks also recorded duplicate ID, image-path, and text-overlap diagnostics.

Figure 5.2: Dataset preparation, image-availability filtering, and final split retention evidence.

The image reliability score is computed as the mean of availability and normalized visual quality cues:

$$q_i = \frac{q_{blur} + q_{brightness} + q_{contrast} + q_{entropy} + q_{available}}{5} \quad (5.1)$$

The text reliability score is computed from length adequacy, truncation quality, and text availability:

$$q_t = \frac{q_{length} + q_{truncation} + q_{available}}{3} \quad (5.2)$$

Chapter 6

Implementation of the Multimodal Fake News Detection Framework

6.1 Introduction

This chapter explains how the adaptive fusion framework was implemented. The implementation used a modular Python repository with separate configuration files, scripts, source modules, tests, notebooks, and output folders. The implementation followed a staged workflow so that data preparation, model training, fusion, analysis, and result export could be verified independently.

6.2 Development Environment

The framework was implemented in Python using PyTorch for deep learning, Transformers for the text branch, scikit-learn for metrics, pandas for tabular processing, and image-processing utilities for visual data handling. Experiments were executed mainly in Google Colab with GPU runtime and Google Drive storage. GitHub was used for version control, and GitHub Desktop was used to manage commits and pushes from the local machine.

The repository was organized into configuration files, scripts, source modules, tests, notebooks, and output folders. YAML configuration files controlled paths, hyperparameters, and experiment variants. This reduced the risk of manually changing code for each experiment.

6.3 Dataset Preparation and Label Standardization

The dataset implementation converted Fakeddit multimodal metadata into a standard project schema. The standard fields were sample identifier, image path, text, and label. Fakeddit two-way labels were converted into the project convention where 0 represents real content and 1 represents fake content.

The preparation stage exported dataset statistics and checked the data for missing required fields, empty text, duplicate identifiers, duplicate text, and duplicate image paths. A final subset was then created for computable experiments. Image downloading was implemented as a separate stage so that unavailable image URLs could be tracked without corrupting the original metadata.

The final image-available subset contained 26,471 samples after image download filtering. This included 18,893 training samples, 3,798 validation samples, and 3,780 test samples.

6.4 6.4 Image Branch Implementation

The image branch was implemented using a ResNet-based classifier. Images were resized and transformed before being passed into the model. The model produced binary probabilities for fake news detection. Training used the training split, while the best checkpoint was selected using validation macro F1.

The image branch produced output CSV files for the train, validation, and test splits. These files contained sample identifiers, labels, image probabilities, predictions, and confidence values. The final image-only test macro F1 was 0.7769.

6.5 6.5 Text Branch Implementation

The text branch was implemented using a DistilBERT-based classifier. Text samples were tokenized and passed through the pretrained language model with a classification head. The branch produced text probabilities, predictions, and confidence values for all splits.

The final text-only test macro F1 was 0.8388. The text branch was stronger than the image branch, but it also showed a larger train-test gap. This behaviour was recorded as an implementation and evaluation limitation rather than hidden.

6.6 6.6 Reliability Feature Extraction

Reliability feature extraction merged image and text outputs and added quality-related features. Image quality was estimated using engineered visual statistics such as blur, contrast, entropy, and related normalized indicators. Text quality was estimated using text-based indicators such as length and availability. The reliability builder also computed confidence difference, quality difference, and modality disagreement.

The output of this stage was a set of reliability-enriched CSV files for train, validation,

and test splits. These files were the direct inputs to baseline evaluation, RL fusion training, supervised fusion, robustness analysis, and explainability.

6.7 6.7 Reinforcement Learning Fusion Implementation

The adaptive fusion controller was implemented as an offline contextual-bandit style reinforcement learning module. For each sample, the controller received a reliability-aware state vector and selected an action from the fusion action space. Each action corresponded to image and text weights. The final probability was computed using the selected weights and the modality probabilities.

The reward was based on classification correctness. A correct fused prediction received positive reward, and an incorrect prediction received negative reward. The model was trained on the training split, selected using validation macro F1, and evaluated on the test split.

The final main RL fusion run achieved 0.8676 test macro F1. The implementation also stored policy analysis files containing action distribution, average modality weights, group-level behaviour, and oracle upper-bound information for the action space.

6.8 6.8 Supervised and Deterministic Baseline Implementation

Several baselines were implemented to evaluate the proposed method. Image-only and text-only baselines used the unimodal branch outputs. Equal fusion averaged image and text probabilities. Confidence-weighted fusion used prediction confidence to weight modalities. Reliability-weighted fusion used reliability indicators in a deterministic fusion rule. A supervised MLP fusion model was also implemented as a learned non-RL comparison.

Threshold tuning was implemented as a validation-based post-processing stage. The threshold was selected using validation macro F1 and then applied unchanged to the test split. This avoided selecting the threshold directly on test data.

6.9 6.9 Explainability Implementation

The explainability implementation generated three types of outputs. Grad-CAM heatmaps were produced for image predictions. Token saliency CSV files were produced for text predictions. Fusion-level explanation summaries recorded selected action, image weight, text weight, final probability, and final prediction.

The final explainability run generated image, text, and fusion-level explanation artifacts. The corresponding summary file was exported under the final metrics directory for traceability.

6.10 6.10 Testing and Reproducibility

The repository includes automated tests for dataset preparation, image inference, text modelling, reliability baselines, RL fusion, ablation, robustness, explainability, supervised fusion, threshold tuning, seed significance, and result export. These tests were used to verify that the implementation stages produced expected files and followed the configured schema.

The six notebooks in the `notebooks` folder provide a reproducible demonstration workflow. They cover data preparation, image model training, text model training, baselines and reliability, RL adaptive fusion, and explainability with final results.

6.11 6.11 Summary

This chapter described the implementation of the proposed framework. The project was implemented as a modular Python pipeline with configurable stages and file-backed outputs. The next chapter evaluates the implemented framework using final experiment results.

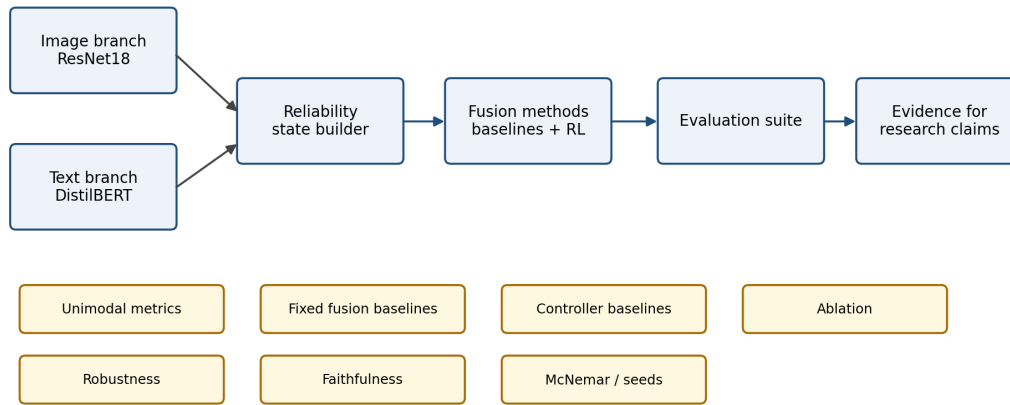


Figure 6.1: Implementation-to-evaluation evidence flow used to connect saved artifacts with final thesis claims.

Chapter 7

Evaluation of the Adaptive Fusion Framework

7.1 Introduction

This chapter evaluates whether the proposed framework achieved the research objectives defined in Chapter 1 and operationalized through the approach in Chapter 4. The evaluation focuses on classification performance, adaptive fusion behaviour, reliability feature contribution, robustness, and explainability. The results are reported cautiously because the final dataset is derived from one benchmark source and the multi-seed evaluation showed variation in reinforcement learning performance.

7.2 Evaluation Strategy

The evaluation was designed around five questions. First, the unimodal image and text branches were evaluated separately to determine the strength of each modality. Second, deterministic fusion baselines were compared with the proposed reinforcement learning fusion model. Third, ablation experiments tested whether reliability-aware state features improved the fusion controller. Fourth, policy and robustness analyses examined whether the model changed modality weights in meaningful ways. Fifth, explainability outputs were inspected to verify that image, text, and fusion-level explanations were generated.

The final evaluation evidence-flow diagram in this chapter makes these tests explicit. It is included to avoid presenting the final macro F1 score in isolation and to show how each examiner-facing question is connected to an artifact-backed test.

7.3 Experimental Setup

The implementation used Python, PyTorch, Transformers, pandas, scikit-learn, and image-processing libraries. Experiments were executed using Google Colab GPU runtime and Google Drive storage. The image branch used a pretrained ResNet-based model, and the

text branch used a DistilBERT-based model. The fusion layer operated on low-dimensional reliability features rather than raw image or text embeddings, which kept the adaptive controller lightweight compared with the unimodal encoders.

7.4 7.4 Dataset and Splits

The experiments used the Fakeddit multimodal dataset. The full prepared metadata contained 682,661 rows after label standardization. For the final computable experiment, 28,000 balanced image-text samples were requested. Due to unavailable or failed image downloads, 26,471 samples were available for final multimodal evaluation. The final available splits contained 18,893 training rows, 3,798 validation rows, and 3,780 test rows.

The original Fakeddit two-way label was converted into the project label standard, where 0 represents real content and 1 represents fake content. Samples without available images were excluded from the final multimodal subset rather than being synthetically filled.

7.5 7.5 Evaluation Metrics

The main classification metrics were macro F1, accuracy, balanced accuracy, precision, recall, confusion matrix, and ROC-AUC. Macro F1 was treated as the primary metric because the task involves two classes and the evaluation should not rely only on overall accuracy. ROC-AUC was used to evaluate ranking quality from predicted probabilities. For policy analysis, average image weight, average text weight, action distribution, and group-wise metrics were also reported.

7.6 7.6 Unimodal Branch Results

The image-only branch achieved a test macro F1 of 0.7769 and test accuracy of 0.7791. The text-only branch achieved a test macro F1 of 0.8388 and test accuracy of 0.8389. These results show that the text branch was stronger than the image branch on the final test split. However, the image branch still contributed useful information because fusion methods improved over the text-only baseline.

The text branch also showed a visible overfitting gap: the training macro F1 was approximately 0.9882, while the test macro F1 was 0.8388. The model checkpoint was selected using validation performance, so the reported test result remains a held-out evaluation, but the gap indicates that stronger regularization or validation-loss-based selection should be considered in future work.

7.7 7.7 Baseline and Fusion Results

The final method comparison is summarized in Table 7.1 and visualized through the performance dashboard, confusion-matrix panel, and macro-F1 comparison figure. Image-only and text-only baselines were weaker than multimodal fusion. Equal fusion and confidence-weighted fusion both achieved test macro F1 of 0.8673. The proposed RL adaptive fusion model achieved test macro F1 of 0.8676 in the main final run. With validation-selected threshold tuning, RL adaptive fusion achieved test macro F1 of 0.8685.

The improvement over the strongest fixed fusion baseline was modest. Therefore, the result should be interpreted as competitive performance with a small main-run improvement, rather than as a large statistical superiority claim. The additional contribution of the proposed method is that it provides sample-level fusion actions and modality weights, which fixed fusion baselines do not provide.

The matched supervised MLP fusion baseline reached 0.8547 macro F1 at the default threshold and 0.8565 macro F1 after threshold tuning. This fairer comparison used the matched supervised fusion evidence files and shows that the adaptive RL fusion result remained stronger in the final reported run.

A paired McNemar audit aligned predictions by sample identifier before testing. The RL adaptive fusion run was not significantly different from equal fusion or confidence-weighted fusion at the default threshold (both $p=1.0000$). Against the matched supervised MLP fusion controller, however, the RL controller produced significantly more correct discordant decisions ($b=95$, $c=46$, $p=0.000045$). This supports the claim that the learned RL controller improved over the matched supervised controller, while fixed-fusion superiority remains a cautious and non-significant claim.

7.8 7.8 Ablation Study

The ablation study tested three reinforcement learning state representations and compared them with simple same-state controller baselines. The probabilities-only state achieved test macro F1 of 0.8523. Adding prediction confidence improved the score to 0.8590. The full reliability state, which included probability, confidence, quality, disagreement, confidence difference, and quality difference, achieved 0.8673. This pattern indicates that the reliability-aware features contributed meaningfully to the fusion controller.

7.9 Policy Analysis and Reliability Behaviour

The final policy analysis, including the weight-behaviour and action-distribution diagrams, showed that the RL fusion model used an average image weight of 0.6594 and an average text weight of 0.3406 on the test split. This indicates that the policy often assigned a larger contribution to the image branch in the selected final run. However, the action distribution also showed that the model did not use only one fixed weighting pattern.

When image and text branch predictions agreed, macro F1 was 0.9110. When they disagreed, macro F1 decreased to 0.7156, and the weights became more balanced, with average image weight 0.4987 and average text weight 0.5013. This behaviour is important because disagreement samples are harder and require more careful fusion decisions.

7.10 Robustness Analysis

Robustness tests were used to examine whether the fusion policy changed modality weights when modality quality or probabilities were corrupted. Under image quality degradation, the model reduced image weight as the degradation level increased, as shown in Figure 7.4. At degradation levels 0.25, 0.50, and 0.75, the image-weight changes were -0.0164, -0.0632, and -0.1283, respectively. Macro F1 decreased only slightly from 0.8671 to 0.8658 across these image-quality degradation levels.

The text-quality degradation behaviour was less consistent. In the tested configuration, the model did not always adapt in the expected direction for text-quality drops. This limitation is discussed in Chapter 8, and it suggests that text reliability estimation requires stronger validation.

7.11 Explainability Evaluation

The explainability component generated qualitative explanation artifacts and a deletion-based faithfulness check. For the image branch, Grad-CAM heatmaps were generated to highlight image regions influencing the prediction. For the text branch, token saliency files were generated to identify influential text tokens. For the fusion decision, the output included selected action, image weight, text weight, final probability, and final prediction.

A larger faithfulness check was conducted on 300 final test samples. Text salient deletion produced a larger probability drop than least-salient and random deletion, which supports the text saliency ranking. Image salient deletion did not outperform random deletion in this metric, so the thesis treats image Grad-CAM as qualitative visual evidence rather than a fully validated faithfulness result.

7.12 7.12 Seed Stability and Statistical Considerations

A small multi-seed check and a controller-baseline seed comparison were conducted for the RL fusion controller. The selected seed 42 run achieved 0.8676 macro F1, compared with 0.8673 for equal fusion. However, the paired McNemar test against equal fusion was not significant ($p=1.0000$), and additional seeds produced lower RL macro F1 than the equal-fusion baseline. Across seeds 42, 7, and 13, RL adaptive fusion achieved mean macro F1 of 0.8594 with a standard deviation of 0.0072. In contrast, the same-state decision-tree controller achieved 0.8529 macro F1, and the RL controller exceeded that controller in all three tested seeds. The thesis therefore avoids claiming statistically proven superiority over fixed fusion. The safer conclusion is that the proposed model demonstrates adaptive and explainable fusion with competitive performance, stronger performance than matched learned controller baselines, and a remaining need for improved seed stability.

7.13 7.13 Discussion of Findings

The evaluation supports the usefulness of multimodal fusion because fusion methods outperformed both image-only and text-only baselines. It also supports the value of reliability-aware state features because the full state outperformed reduced ablation settings. The adaptive fusion model provided interpretable modality weights and action records, which directly addressed the explainability aspect of the research problem.

At the same time, the performance difference between RL adaptive fusion and strong fixed fusion was small. This suggests that the final data split contains strong complementarity between image and text predictions that can already be captured by simple fusion. The proposed framework is therefore best defended as an explainable adaptive fusion framework with competitive performance and modest main-run difference, not as a method that universally dominates fixed fusion.

7.14 7.14 External Generalization Case Study

An additional external case study was conducted using the OpenFake Reddit test split. This dataset was not used for training or model selection. A balanced external sample of 3,780 image-text posts was exported, matching the scale of the final Fakeddit test split. The purpose of this experiment was to examine whether the Fakeddit-trained models transferred directly to a different multimodal data source.

The zero-shot OpenFake result showed strong domain shift. The image-only branch achieved weak external performance, and the text branch assigned very high fake-class

probabilities to almost all samples at the default 0.5 threshold. However, the text branch retained ranking signal: after selecting a threshold on 20 percent of OpenFake and applying it unchanged to the remaining 80 percent holdout subset, text-only performance reached 0.8589 macro F1. In contrast, equal fusion, reliability-weighted fusion, and RL adaptive fusion remained near 0.49 macro F1 after the same calibration protocol.

This finding is important because it separates ranking transfer from calibration transfer. The DistilBERT text representation carried transferable signal, but its probabilities were poorly calibrated under the OpenFake distribution. The image branch and the Fakeddit-trained fusion controller did not transfer well in this zero-shot setting. Therefore, external deployment would require calibration, domain adaptation, or additional external training data.

7.15 7.15 Summary

This chapter evaluated the proposed framework using final dataset statistics, unimodal results, fusion comparisons, ablation, policy analysis, robustness tests, explainability artifacts, and seed stability checks. The results show that the framework achieved competitive performance and provided sample-level adaptive explanations. The next chapter concludes the thesis by summarizing contributions, objective achievement, limitations, and future work.

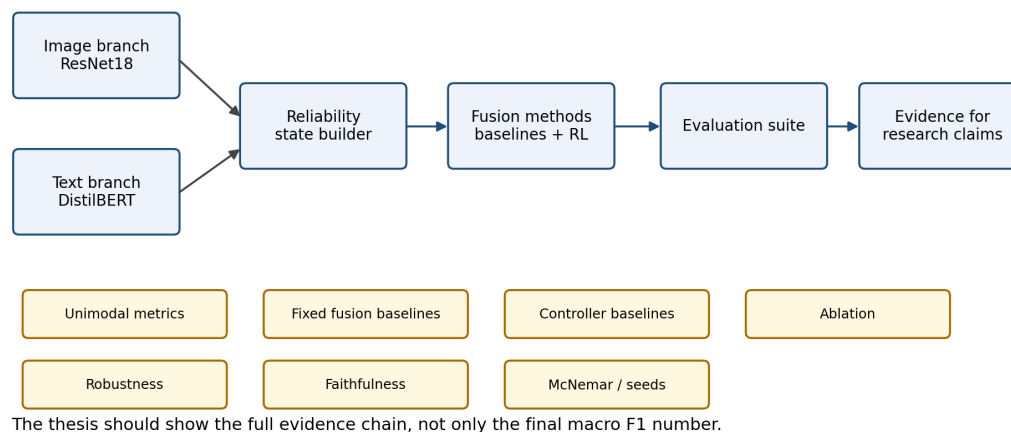


Figure 7.1: Final evaluation evidence flow across unimodal models, fusion methods, robustness, explainability, and statistical checks.

Table 7.1: Dataset preparation and retention evidence for the final multimodal experiment.

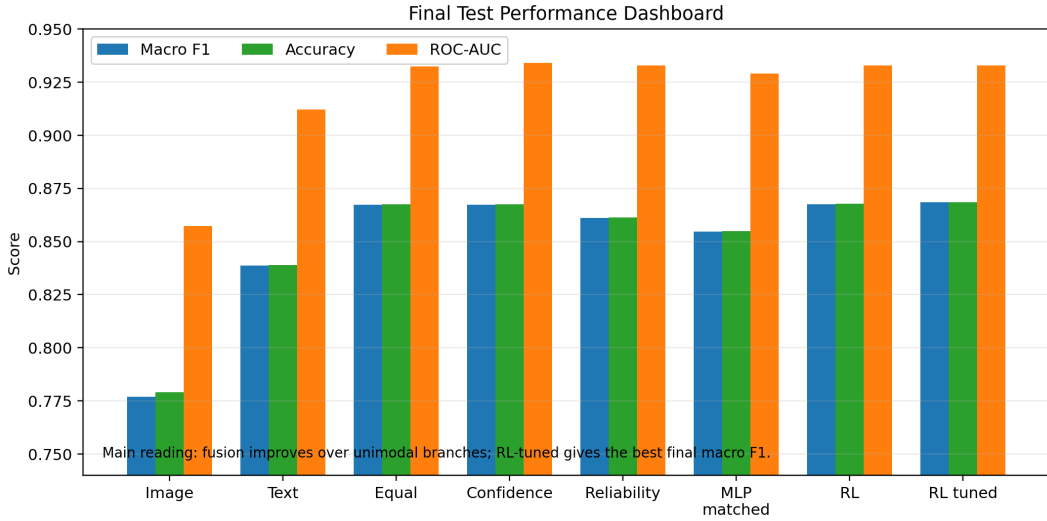
stage	requested_or_source	available	failed_or_removed	evidence
Full prepared metadata	682661	682661	0	fakeddit-stats.json
Final requested subset	28000	26471	1529 failed image downloads	final image download summary
Train split	20000 requested	18893	1107 failed	final_splits_available/train.csv
Validation split	4000 requested	3798	202 failed	final_splits_available/validation.csv
Test split	4000 requested	3780	220 failed	final_splits_available/test.csv

Macro F1 was used as the primary classification metric because it gives equal importance to both classes:

$$F1_{macro} = \frac{1}{C} \sum_{c=1}^C F1_c \quad (7.1)$$

Validation-threshold tuning selected the threshold that maximized validation macro F1 and then applied that threshold unchanged to the test split:

$$\tau^* = \arg \max_{\tau \in T} F1_{macro}^{val}(\tau) \quad (7.2)$$

**Figure 7.2:** Final test performance dashboard across accuracy, macro F1, and ROC-AUC.

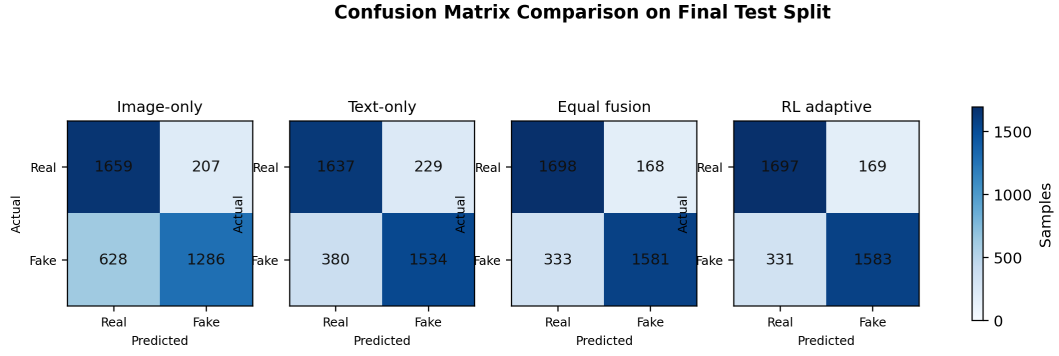


Figure 7.3: Confusion matrix comparison for unimodal, fixed-fusion, and adaptive-fusion methods.

Final Test Macro F1 Comparison

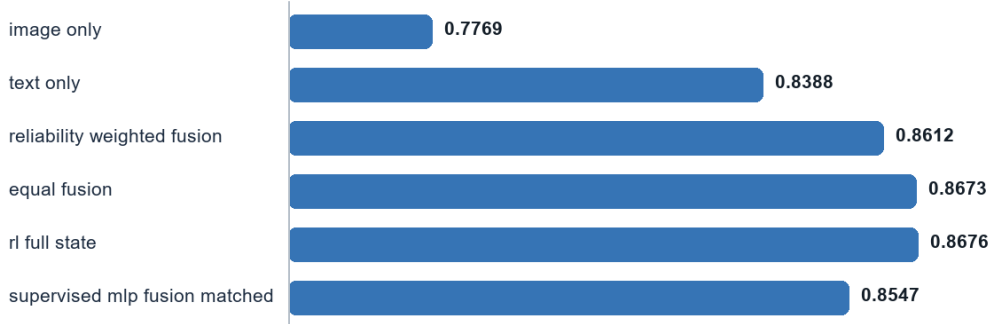


Figure 7.4: Final test macro F1 comparison across selected methods.

Table 7.2: Final test method comparison.

source	method	macro.f1	accuracy	roc.auc
rl	rl_full.state	0.8676044474818365	0.8677248677248677	0.9328939130746426
baseline	equal_fusion	0.8673332160630083	0.8674603174603175	0.932541682486244
baseline	confidence_weighted_fusion	0.8673332160630083	0.8674603174603175	0.9339962436203704
ablation	rl_full.reliability.state	0.8673332160630083	0.8674603174603175	0.9326433197704956
baseline	reliability_weighted_fusion	0.8612064094284793	0.8613756613756614	0.9328661938153012
ablation	rl_prediction.confidence	0.8589942254358114	0.858994708994709	0.9184258596610296
supervised_fusion	supervised_mlp.fusion_matched	0.854718945966428	0.8547619047619047	0.9291012744139476
ablation	rl_probabilities.only	0.8523148018484805	0.8523809523809524	0.9205120839171176
supervised_fusion	supervised_mlp.fusion	0.8499976379068706	0.85	0.9290694952630866
baseline	text_only	0.838769176548389	0.8388888888888889	0.912095228815486
baseline	image_only	0.7769284368078044	0.7791005291005291	0.8572159951886086

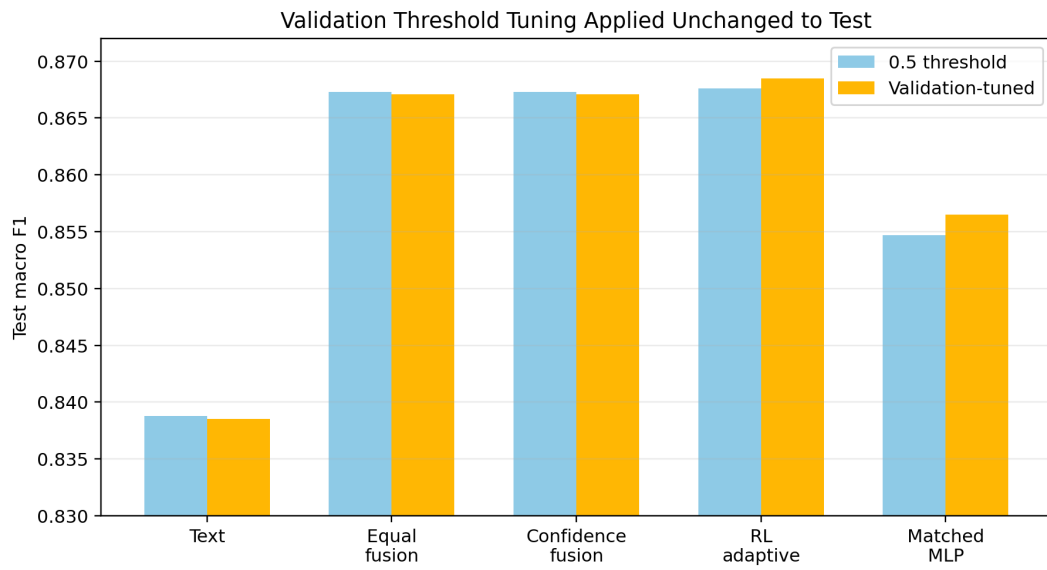


Figure 7.5: Comparison of default-threshold and validation-tuned threshold results.

Ablation Results by Fusion State Representation



Figure 7.6: Ablation results for different RL fusion state representations.

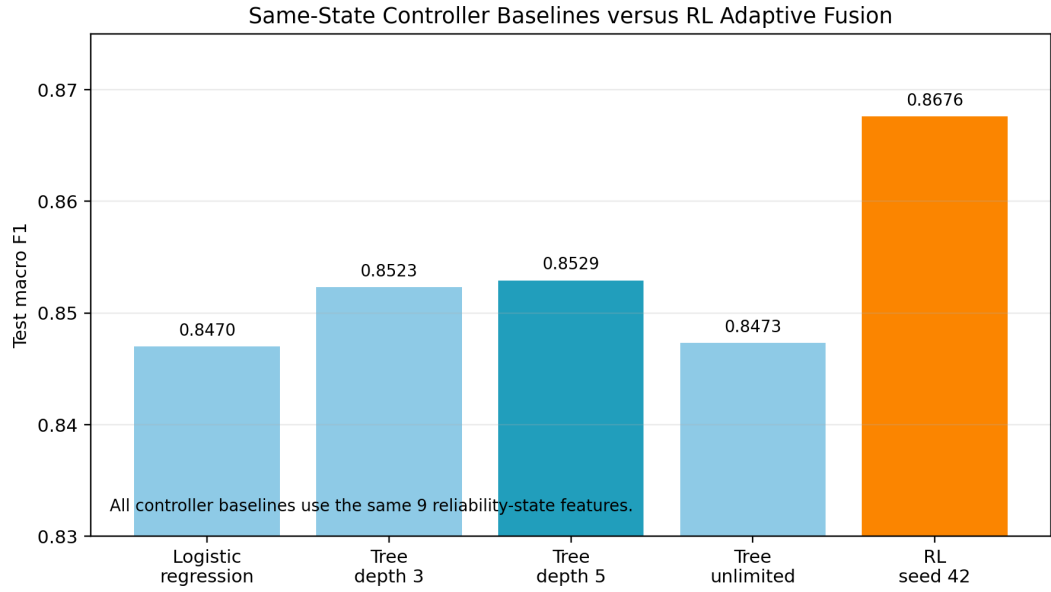


Figure 7.7: Same-state controller baseline comparison against the RL adaptive fusion controller.

Table 7.3: Same-state controller baseline results.

method	state_dim	validation_macro_f1	test_macro_f1	test_accuracy
state_logistic_regression	9	0.8541	0.8470	0.8471
state_decision_tree_depth_3	9	0.8576	0.8523	0.8524
state_decision_tree_depth_5	9	0.8604	0.8529	0.8529
state_decision_tree_unlimited	9	0.8541	0.8473	0.8474
rl_adaptive_fusion_seed_42	9	0.8678	0.8676	0.8677

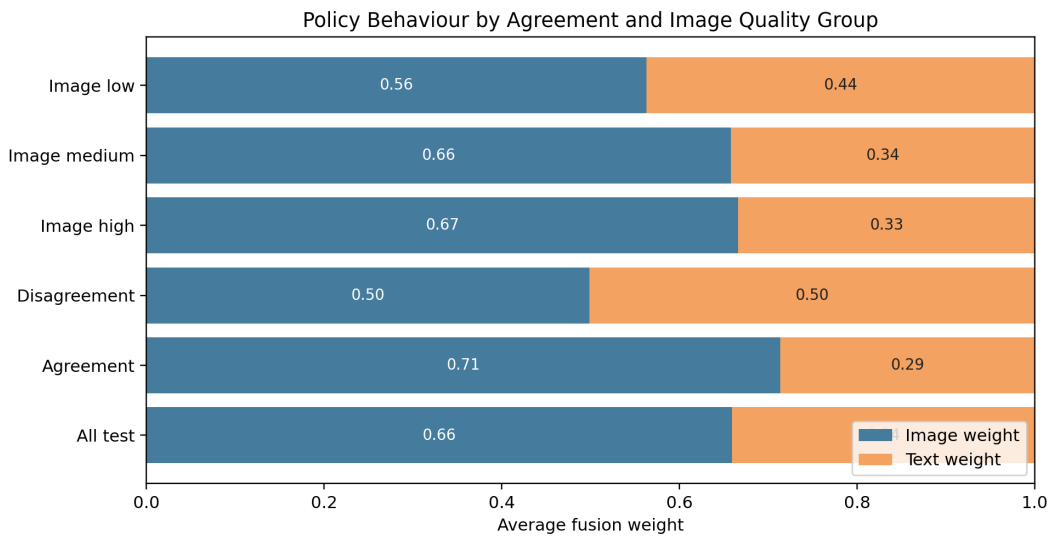


Figure 7.8: Average modality weights selected by the RL policy across agreement and image-quality groups.

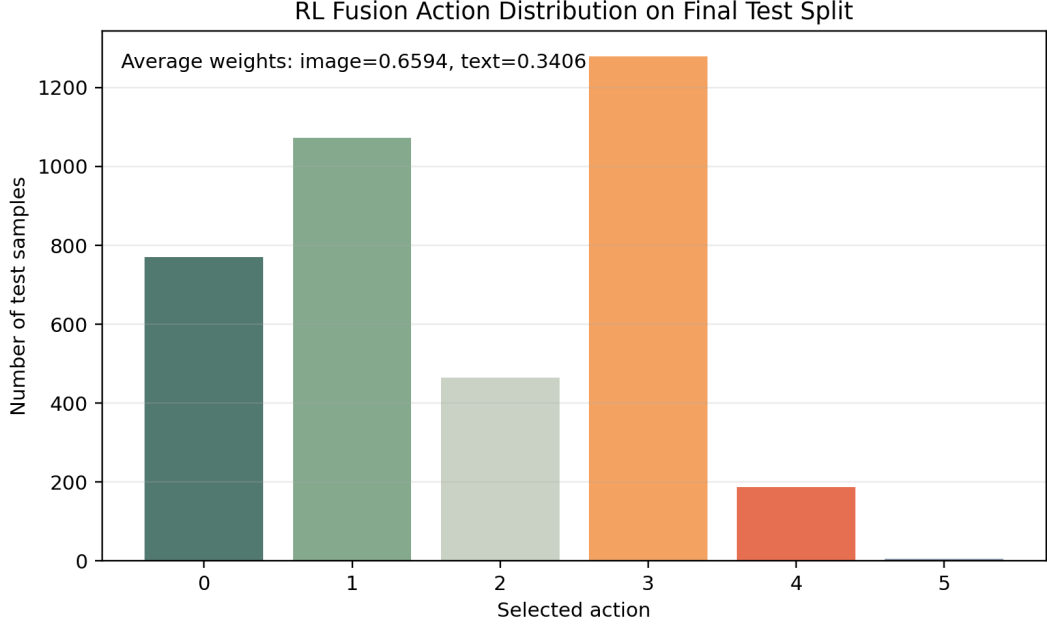


Figure 7.9: Distribution of selected RL fusion actions on the final test split.

Image Quality Degradation and Image Weight Adaptation

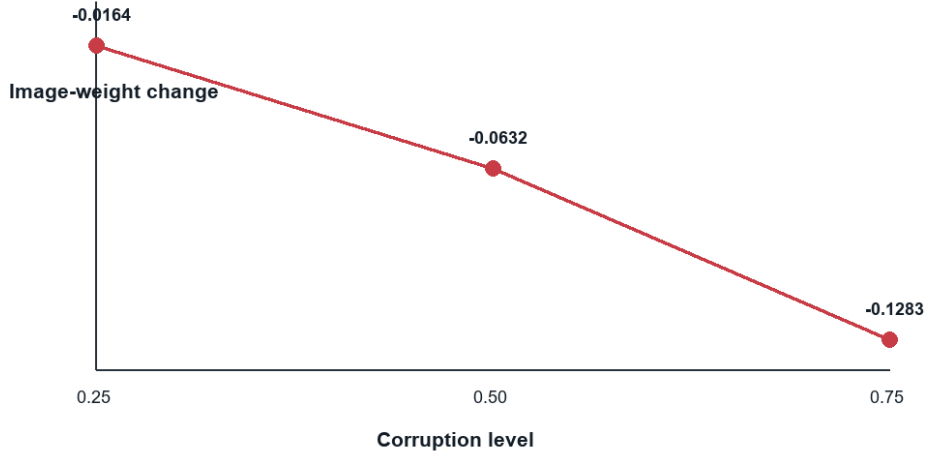


Figure 7.10: Image quality degradation and image-weight adaptation.

Deletion-based faithfulness was measured using comprehensiveness, defined as the probability drop after removing evidence E :

$$Comp(E) = P(y_c | x) - P(y_c | x_{\setminus E}) \quad (7.3)$$

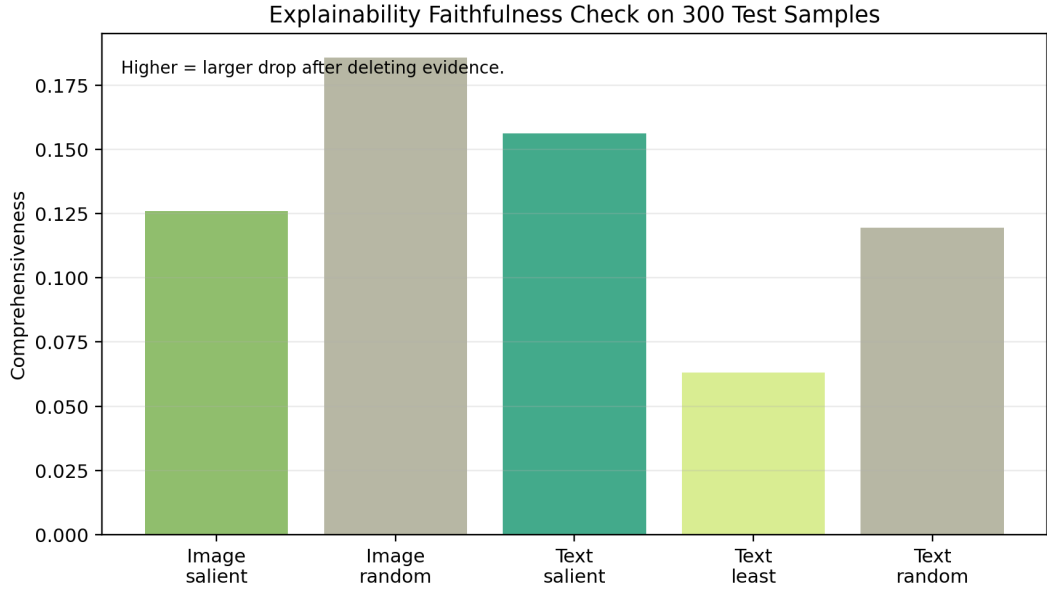


Figure 7.11: Deletion-based explainability faithfulness check on 300 final test samples.

Table 7.4: Explainability faithfulness summary for image and text evidence deletion.

explanation_test	samples	salient	control	margin	finding
image Grad-CAM deletion	300	0.1260	random 0.1858	-0.0598	not passed
text saliency vs least-salient deletion	300	0.1562	0.0631	0.0931	passed
text saliency vs random deletion	300	0.1562	0.1195	0.0367	passed

For paired classifier comparison, McNemar’s test uses the discordant correctness counts:

$$b = |\{j : \hat{y}_j^A = y_j, \hat{y}_j^B \neq y_j\}|, \quad c = |\{j : \hat{y}_j^A \neq y_j, \hat{y}_j^B = y_j\}| \quad (7.4)$$

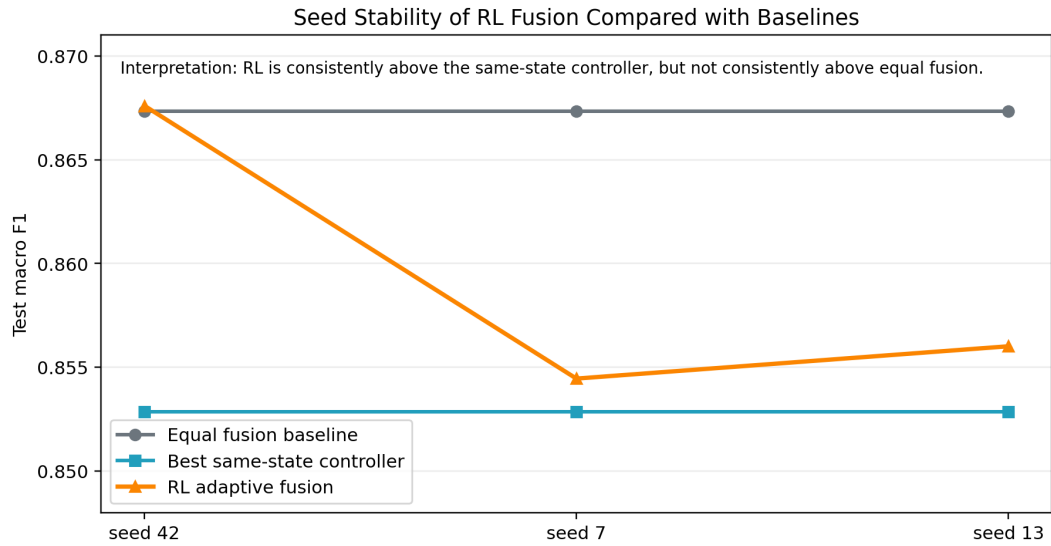


Figure 7.12: Seed-stability comparison between RL adaptive fusion, equal fusion, and the best same-state controller baseline.

Table 7.5: External OpenFake calibration-holdout case study results.

Method	Selected threshold	Holdout macro F1	Holdout accuracy	Holdout ROC-AUC
text_only	0.991	0.858926	0.859127	0.859127
rl_adaptive_fusion	0.889	0.489392	0.490410	0.474842
equal_fusion	0.894	0.484248	0.485450	0.478006
image_only	0.788	0.480838	0.481812	0.468784

Table 7.6: Research-question and examiner-question evidence mapping.

examiner_question	test_or_evidence	main_finding
Novelty and architecture difference	adaptive fusion architecture; same-state controller baselines	sample-level RL action and weights provide inspectable fusion
Baseline comparison	image/text/equal/confidence/reliability/MLP/RL comparisons	RL competitive; threshold-tuned RL macro F1 0.8685
Why reliability features	state ablation	full reliability state outperformed probability-only and confidence states
Dataset filtering	download and validation statistics	26,471 final multimodal rows retained; failures recorded
Generalizability	threats-to-validity analysis	claims restricted to Fakeddit; cross-dataset work remains future work
Explainability	Grad-CAM, token saliency, fusion weights, N=300 faithfulness	text saliency positive; image faithfulness remains limited

Chapter 8

Conclusion and Further Work

8.1 Introduction

This chapter concludes the thesis by summarizing the research contributions, objective achievement, quantitative results, limitations, challenges, and future work. The conclusion is written in relation to the research aim: to develop and evaluate an explainable multimodal fake news detection framework that adaptively fuses image and text predictions using reinforcement learning.

8.2 Summary of Research Contributions

The main contribution of this research is an explainable adaptive fusion framework for multimodal fake news detection. The framework combines a ResNet-based image branch, a DistilBERT-based text branch, reliability feature extraction, reinforcement learning-based fusion, baseline evaluation, robustness analysis, and explanation generation.

The first contribution is the formulation of image-text fusion as a reliability-aware action-selection problem. Instead of using a fixed fusion rule for all samples, the system observes a state containing prediction, confidence, quality, and disagreement features, then selects a fusion action.

The second contribution is the implementation of a reproducible experimental pipeline. The project includes dataset preparation, image downloading, model training, inference, reliability computation, baseline comparison, RL fusion, supervised fusion, ablation, robustness, threshold tuning, seed checks, and explainability outputs.

The third contribution is the explanation mechanism. The framework produces Grad-CAM image explanations, token-level text saliency, and fusion-level action/weight explanations.

8.3 8.3 Achievement of Objectives

The first objective was to conduct a critical literature review. This was addressed by reviewing fake news detection, multimodal learning, fusion strategies, reinforcement learning, and explainable AI. The review identified the gap for sample-level adaptive and explainable fusion.

The second objective was to study relevant technologies and theoretical foundations. This was addressed through the study of transformer-based text modelling, residual image modelling, reinforcement learning, reliability estimation, and explanation methods.

The third objective was to design a reliability-aware adaptive fusion approach. This was achieved by defining a state representation, action space, reward design, and fusion workflow.

The fourth objective was to implement a reproducible framework. This was achieved through the project repository, notebooks, configuration files, source code modules, and output evidence files.

The fifth objective was to evaluate the framework. This was achieved through held-out test evaluation, baseline comparison, ablation, robustness analysis, explainability examples, and seed stability checks.

8.4 8.4 Quantitative Performance Summary

The final evaluation used 26,471 available multimodal samples from the Fakeddit-derived final subset. The image-only branch achieved 0.7769 test macro F1, while the text-only branch achieved 0.8388. The best fixed fusion baselines achieved approximately 0.8673 test macro F1. The main RL adaptive fusion run achieved 0.8676 test macro F1, and validation-selected threshold tuning improved it to 0.8685.

The ablation study showed that the full reliability-aware state improved over reduced state representations. Probabilities alone achieved 0.8523 test macro F1, prediction and confidence achieved 0.8590, and the full reliability state achieved 0.8673. This supports the usefulness of reliability-aware state features.

The matched supervised MLP fusion baseline achieved 0.8547 macro F1 at the default threshold and 0.8565 after threshold tuning. The RL fusion result was higher in the main final run, and paired McNemar testing showed a significant difference against the matched supervised MLP controller ($p=0.000045$). However, multi-seed checks showed that the RL improvement over fixed fusion was not stable across all tested seeds. Therefore, the performance conclusion is stated cautiously.

8.5 8.5 Limitations of the Study

The first limitation is dataset scope. The final evaluation used a subset derived from one dataset source. Therefore, the main results demonstrate within-dataset generalization, not universal real-world generalization. An additional OpenFake external case study showed that cross-dataset transfer was difficult: the text branch retained ranking signal after calibration, but the image branch and the Fakeddit-trained fusion policy did not transfer strongly in zero-shot form.

The second limitation is seed stability. The main RL run achieved a modest improvement, but additional seeds showed weaker RL performance. This indicates that the controller is sensitive to training conditions.

The third limitation is text-branch overfitting. The text branch showed a large train-test performance gap. Although validation-based checkpoint selection was used, stronger regularization should be investigated.

The fourth limitation is explainability validation. A deletion-based faithfulness check was conducted on 300 samples, but the results were mixed: text token saliency showed positive faithfulness, while image Grad-CAM did not outperform random image-region deletion in this metric. Therefore, image explanations should be treated as qualitative visual evidence rather than fully validated causal explanations.

The fifth limitation is the simplicity of the reward function. The binary reward supports clear training but does not reward calibration or probability margin.

8.6 8.6 Challenges Encountered

Several practical challenges were encountered during the research. Image downloading from the dataset was incomplete because some external image URLs were unavailable. Computational limitations required the use of manageable final subsets and GPU runtime planning. The fusion controller required careful baseline comparison because small numerical improvements can be misleading if training budgets are not controlled. Finally, explainability outputs had to be designed so that image, text, and fusion-level evidence could be inspected together.

8.7 8.7 Future Work

Future work should evaluate the framework on additional multimodal fake news datasets and cross-domain event splits. The OpenFake case study indicates that future versions should include external-domain calibration and domain adaptation rather than assuming

that probabilities learned from one dataset will transfer unchanged. The RL fusion controller should also be trained with more seeds, tuned more systematically, and compared using statistical tests on larger evaluation protocols.

The text branch could be improved using stronger regularization, validation-loss-aware checkpoint selection, or domain-specific pretraining. The reward function could be extended from binary correctness to margin-based or log-loss-based rewards. The action space could also be expanded or converted into a continuous policy if higher fusion precision is required.

Finally, explainability evaluation should be expanded beyond the current deletion-based check using insertion tests, larger perturbation protocols, and human-centered studies to determine whether the generated explanations are useful and trustworthy.

8.8 8.8 Summary

This thesis presented an explainable reinforcement learning-based adaptive fusion framework for multimodal fake news detection. The framework achieved competitive performance against strong baselines and provided sample-level fusion explanations through modality weights and saliency outputs. The research demonstrates that adaptive fusion is a useful direction for explainable fake news detection, particularly when compared with matched learned controller baselines, while also identifying stability, generalizability, and expanded explainability validation as important future work.

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Appendix A: Reproducibility and Artifact Manifest

This appendix lists the main reproducibility artifacts for the project.

.1 Notebooks

- `notebooks/01_data_preparation.ipynb`: dataset preparation and split creation.
- `notebooks/02_image_model_training.ipynb`: image model training and inference workflow.
- `notebooks/03_text_model_training.ipynb`: text model training and inference workflow.
- `notebooks/04_baselines_and_reliability.ipynb`: baseline fusion and reliability feature workflow.
- `notebooks/05_rl_adaptive_fusion.ipynb`: reinforcement learning fusion training and evaluation.
- `notebooks/06_explainability_and_final_results.ipynb`: explainability generation and final result export.

.2 Main Configurations

- `configs/final_subset.yaml`
- `configs/final_image_download.yaml`
- `configs/final_image_model.yaml`
- `configs/final_text_model.yaml`
- `configs/final_reliability.yaml`
- `configs/final_baselines.yaml`
- `configs/final_fusion.yaml`
- `configs/final_supervised_fusion_matched.yaml`
- `configs/final_threshold_tuning_matched.yaml`
- `configs/final_rl_analysis.yaml`
- `configs/final_robustness.yaml`
- `configs/final_explainability.yaml`
- `configs/final_results_export.yaml`

.3 Main Results

- `outputs/tables/final_method_comparison.csv`
- `outputs/tables/final_results_summary.json`
- `outputs/tables/final_stage_status.csv`
- `outputs/metrics/final_threshold_tuning_matched_summary.csv`
- `outputs/metrics/final_seed_significance_summary.csv`
- `outputs/metrics/final_explainability_summary.json`
- `outputs/robustness/final_robustness_summary.csv`